

user authentication and security, Polly for text to speech conversion for conversational logs (typed in a chatbox), and DynamoDB stores and retrieves the data logs for chat service.

Though a similar chatbot service can be developed with Azure or GCP, the AWS CloudFormation template in the Serverless Bot Framework makes it easy and simple to develop, customise and deploy a bot service on the fly. We can also extend this architecture to include Amazon Lex, which is a specialised service for conversational chatbots to offer additional facilities like sending an email with standard operating procedure (SOP), creating an incident ticket with backend services, etc.

Alexa Voice Service

Alexa Voice Service (AVS) can be used to trigger a function call in the AWS cloud service such as a Lambda call using the Alexa Skills kit. This Lambda call can be used to trigger CI commands like Jenkins API to enable build using Maven or AWS CodeBuild, test using Junit, and deploy the code build using AWS CLI or AWS CodeDeploy. Finally, the build code can be deployed to a container service like Elastic Beanstalk container, as shown in Figure 4.

This service uses voice commands to trigger build and deployment without the need for manual build commands using AWS CodeStar or similar tools. This can be done from any remote location using integrated Alexa devices connected through the Alexa Voice Service (AVS).

Amazon Fraud Detector

AWS has more than 20 years of expertise in fraud detection. The ML based Amazon Fraud Detector is a fully managed service for processing the event data fed by users for identifying potential frauds. It is especially useful in FinTech applications such as insurance claims, tax filing, bank transactions, credit card transactions, credit score reports, and so on.

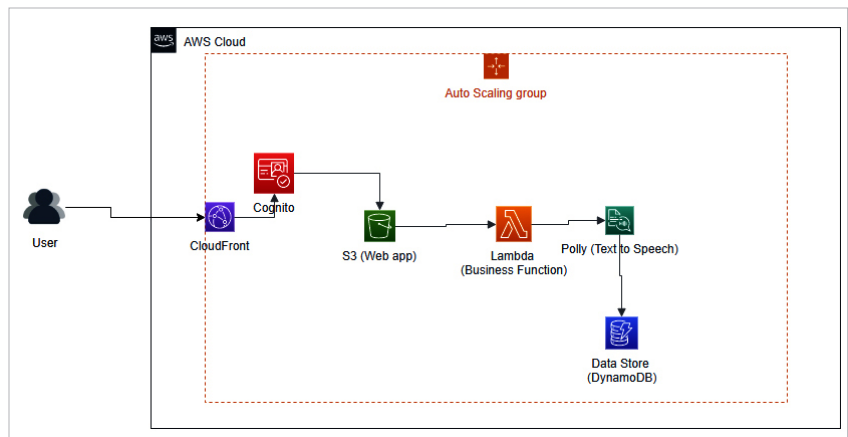


Figure 3: Example implementation of Serverless Bot Framework architecture

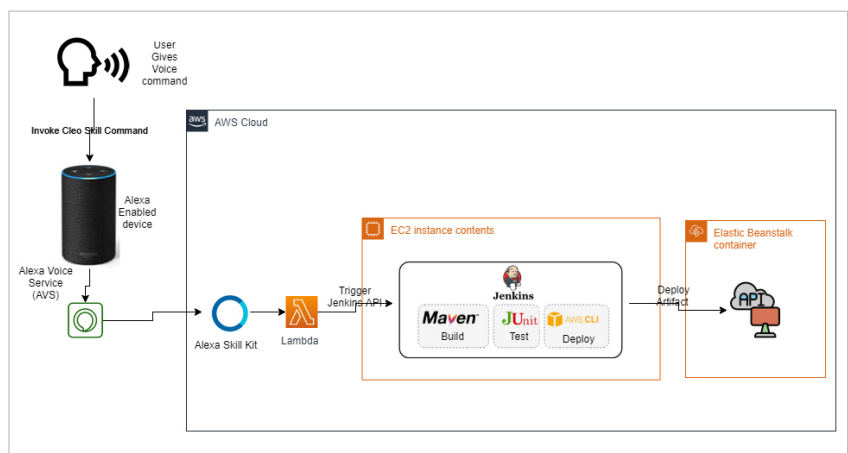


Figure 4: Alexa Voice Service for handling DevOps tasks

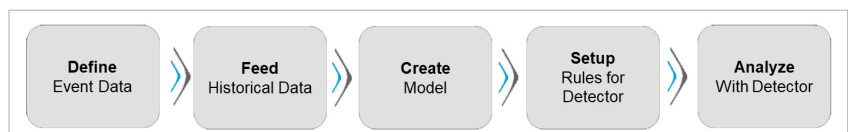


Figure 5: The five stages of execution of AWS Fraud Detector

AWS Fraud Detector has five stages of execution (Figure 5). The user feeds the event data to be processed (e.g., insurance claim) as well as the historical data that will be used for training the model. Once the model is trained, rules are set up to prepare the detector, and finally the Amazon Detector takes these as inputs (training model and rules) and uses 'Detection' logic to provide 'Prediction' results.

Lambda can also be used to trigger SageMaker based machine learning solutions to handle anomaly detection

and fraud detection. But the advantage of using Fraud Detector is that it has in-built data sets for training the model easily. Template models are also available for common use cases like credit card transaction validation, credit score report validation, etc.

Amazon Rekognition

Amazon Rekognition is an advanced image and video analysis tool based on deep learning technology that can help with face recognition, object identification, text extraction, activity